Where Should the LoRA Rank Go?
Layer-wise Capacity Placement under Sequential Fine-Tuning

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Abstract

Low-rank adaptation normally assigns one rank to every adapted Transformer block. We isolate where that capacity is placed by sequentially adapting a 36-block, 4B instruction model to instruction following and mathematics in both orders. Uniform, early-heavy, middle-heavy, and late-heavy allocations all use exactly 5,898,240 trainable LoRA parameters. In one controlled seed, moving the high-rank region deeper increases IFEval stage-to-stage forgetting from 1.85 to 3.14 points among non-uniform allocations, whereas math-first backward transfer ranges only from -0.45 to $+0.30$. The middle-heavy run is the sole protocol to finish above the unadapted checkpoint on either benchmark ($+0.30$ GSM8K), yet 15 of 16 final benchmark cells decline. Thus rank location changes an order-dependent acquisition–retention trade-off, but the evidence does not support a task-independent layer preference.

1. Introduction and Related Work

Sequential fine-tuning can overwrite capabilities acquired earlier, the classical catastrophic-interference problem (McCloskey & Cohen, 1989). LoRA freezes a weight matrix W_l and learns $\Delta W_l = B_l A_l$ with rank r_l (Hu et al., 2022); QLoRA makes this memory-efficient through a frozen 4-bit backbone (Dettmers et al., 2023). The usual constant r_l assigns equal nominal adaptation capacity at every depth. AdaLoRA instead prunes low-importance singular directions (Zhang et al., 2023), while dEBORA learns layer-wise ranks through bilevel optimization (Zangrando et al., 2025). These methods motivate non-uniform capacity, but do not establish how its depth affects backward transfer in sequential LLM adaptation.

We ask two questions: *Do early, middle, and late blocks respond differently when given more rank? Can non-uniform rank improve new-task learning without increasing forgetting?* Our controlled study holds data, token budget, LoRA scaling, and parameter count fixed. It contributes (i) an exact-budget comparison in both task orders, (ii) stage-wise acquisition and backward-transfer measurements, and (iii) a reproducible spectral allocator whose pilot outcome is reported separately, without claiming an unevaluated benchmark result.

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PCS5022 – Neural Networks and Deep Learning

2. Problem and Experimental Design

Let $\mathcal{M}_l$ be the adapted projections in block l. For rank vector $\mathbf{r} = (r_0, \ldots, r_{L-1})$, its exact capacity cost is

$$C(\mathbf{r}) = \sum_{l=0}^{L-1} \sum_{W \in \mathcal{M}_l} r_l \, [m(W) + n(W)]. \quad (1)$$

We compare vectors satisfying $C(\mathbf{r}) = C(16\mathbf{1})$ and set $\alpha_l = r_l$, so $\alpha_l / r_l = 1$. If S_i^t is benchmark i after stage $t \in \{0, 1, 2\}$, we follow Lopez-Paz & Ranzato (2017) and define first-task backward transfer as $\mathrm{BWT}_1 = S_1^2 - S_1^1$ (negative values indicate forgetting). Final net performance is $N_i = S_i^2 - S_i^0$. We report each task separately and define the descriptive utility $U = \mathrm{mean}_i(N_i) - \mathrm{mean}_i(\max_t S_i^t - S_i^2)$ only as a secondary summary.

The starting checkpoint is Qwen3-4B-Instruct-2507 (Qwen Team, 2025), not an untrained base LM. LoRA targets only `q_proj` and `v_proj`. The 36 blocks form three contiguous groups; ranks are $(16, 16, 16)$ or a permutation of $(32, 8, 8)$. The same adapter learns IFEval-like instruction data then NuminaMath, or the reverse, with optimizer state reset at the task boundary. We evaluate stages 0–2 on all 541 IFEval prompts (strict prompt accuracy) and 1,319 GSM8K test problems (numeric exact match) (Zhou et al., 2023; Cobbe et al., 2021). Appendix A gives the full protocol and contamination controls.

3. Results and Discussion

The unadapted checkpoint scores 80.04 on IFEval and 88.10 on GSM8K. Figure 1b shows negative first-task acquisition for all eight protocols. Across both tasks and or-

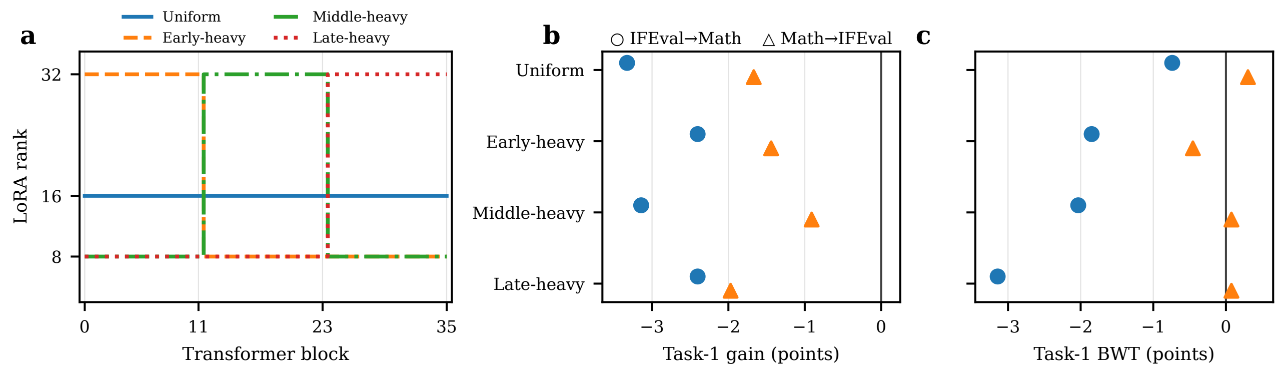

Figure 1. Fixed-budget allocations and their sequential behavior (seed 42). **(a)** Rank by depth; each pattern has 5,898,240 parameters. **(b)** Task-1 score immediately after learning it, relative to the unadapted checkpoint. **(c)** Task-1 backward transfer after task 2; values left of zero indicate forgetting. Rank placement changes retention, but the direction and magnitude depend on task order.

ders, 15 of 16 final cells remain below the starting checkpoint, despite lower training-derived calibration NLL in every run. The experiment therefore measures benchmark-level negative acquisition followed by sequential interference; a final loss should not automatically be called forgetting.

Rank location changes retention. With IFEval first, uniform ranks lose only 0.74 points from stage 1 to stage 2. Among non-uniform patterns, forgetting rises from 1.85 (early-heavy) through 2.03 (middle-heavy) to 3.14 (late-heavy). With mathematics first, the pattern changes: uniform gains 0.30 points after instruction training, middle- and late-heavy gain 0.08, and early-heavy loses 0.45. The effect is therefore strongly order-dependent rather than a fixed preference for one depth region.

Non-uniform ranks yield task-specific, not universal, advantages. For IFEval→Math, middle-heavy is the only run with positive final GSM8K net performance (+0.30), while uniform preserves the highest final IFEval score (75.97). In the reverse order, late-heavy gives the best final IFEval score (78.19), whereas middle-heavy gives the best GSM8K retention (87.26). Early-heavy has the best mean benchmark utility (-4.43 versus -5.34 for late-heavy), while a benchmark-free calibration rule selects Middle-heavy among the three non-uniform candidates. The exact table and Pareto views are in Appendix B. Several differences correspond to only a few benchmark items, and one seed provides no estimate of training variance.

4. Conclusion

Under an equal LoRA budget, where rank is placed changes sequential interference, but no allocation dominates both tasks and orders. Uniform rank best protects instruction following when learned first; middle and late capacity can improve different endpoints in the reverse order. Selection should remain order-aware and benchmark-independent, although whether calibration NLL predicts external retention remains unresolved. Multi-seed and higher-precision controls are required before interpreting these descriptive effects as stable layer preferences.

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A. Protocol and Reproducibility

Model and optimization. We use `Qwen/Qwen3-4B-Instruct-2507` at revision `cdbee75f17c01a7cc42f958dc650907174af0554`. The frozen backbone uses 4-bit NF4 quantization with double quantization and BF16 computation; training uses gradient checkpointing and SDPA. Conventional LoRA is applied to 72 modules (`q_proj` and `v_proj` in 36 blocks), with no RSLoRA, dropout 0.05, and module-wise scaling exactly one. Every allocation contains 5,898,240 trainable parameters (0.147% of the 4.02B parameter model), verified after PEFT instantiation. Four DDP processes each hold one quantized replica. The effective batch size is 16, maximum sequence length 2,048, learning rate 2×10^{-4} with cosine decay and 3% warm-up, and weight decay zero. Optimizer and scheduler are reinitialized at stage 2 while the stage-1 adapter weights are loaded as trainable parameters.

Data and token control. Instruction training uses the `filtered` configuration of `argilla/ifeval-like-data` (Argilla, 2024); mathematics uses `AI-MO/NuminaMath-CoT` (Li et al., 2024). Each stage's unique prepared split contains exactly 1,000,000 supervised assistant tokens; calibration contains 50,000 per task. The official tokenizer chat template is used, and labels outside the assistant answer are -100. Packing preserves these labels. The budgets draw from 7,036 instruction and 2,302 math source examples, packed into 847 and 707 training sequences, respectively.

Decontamination precedes splitting. Text is Unicode-NFKC normalized and checked by SHA-256 exact hashes, MinHash candidates over character 5-grams, and confirmed similarity at 0.90. From 56,339 IFEval-like rows, 4,741 internal duplicates are removed, leaving 51,598; no official IFEval prompt remains. From 859,494 NuminaMath rows, the pipeline exclusively removes 7,342 rows by GSM8K source, 7,662 exact benchmark matches, 847 approximate matches, and 57,618 internal duplicates, leaving 786,025. A final audit finds zero exact overlap with the 541 IFEval prompts and 8,792 GSM8K train-plus-test prompts. Fingerprint-addressed preprocessing directories store removed-item hashes and token manifests; each resolved run configuration links the corresponding model, dataset, and preprocessing revisions.

Evaluation and integrity. Generation is deterministic (greedy decoding) with separate output limits for the two benchmarks. IFEval uses the official verifiers distributed with `lm-eval==0.4.12`; GSM8K retains raw generations and applies a tested numeric final-answer extractor. All 17 distinct evaluation points (one cached starting checkpoint and two stages for eight protocols) contain the complete benchmarks. All 16 adapters are saved as safetensors and were reloaded for evaluation. Each protocol trained on four GPUs and took approximately 14 minutes for the two training stages, excluding generation-based evaluation.

Table 1. Manual rank patterns. Early, middle, and late each contain 12 blocks. Exact parameter equality follows from identical projection shapes.

Allocation	Early	Middle	Late	Parameters
Uniform	16	16	16	5,898,240
Early-heavy	32	8	8	5,898,240
Middle-heavy	8	32	8	5,898,240
Late-heavy	8	8	32	5,898,240

B. Complete Stage-wise Results

The manual-selection utility uses only decontaminated calibration NLL. Among the predeclared non-uniform candidates it selects middle-heavy: mean utilities are 14.439 (early), 14.469 (middle), and 14.440 (late). Uniform is the experimental reference and was excluded from that selection; had it been eligible, its mean calibration utility of 14.507 would be largest. No IFEval or GSM8K score participates in selection.

At the benchmark level, mean descriptive utility across orders is -4.426 (early), -4.500 (middle), -5.044 (uniform), and -5.343 (late). Because all configurations have the same parameter count, no performance–parameter curve can be identified; Figure 4 displays the controlled vertical slice rather than introducing artificial jitter.

Table 2. All benchmark scores (0–100). I and M denote instruction and mathematics. $\mathrm{BWT}_1 = S_1^2 - S_1^1$; positive values indicate that training task 2 improved the first-task benchmark. Boldface is intentionally omitted because the best endpoint depends on the comparison of interest.

Allocation	Order	IFEval			GSM8K			BWT_1
		Base	S1	S2	Base	S1	S2	
Uniform	I→M	80.04	76.71	75.97	88.10	87.87	87.41	-0.74
Uniform	M→I	80.04	80.22	76.16	88.10	86.43	86.73	+0.30
Early-heavy	I→M	80.04	77.63	75.79	88.10	88.10	87.79	-1.85
Early-heavy	M→I	80.04	79.85	77.63	88.10	86.66	86.20	-0.45
Middle-heavy	I→M	80.04	76.89	74.86	88.10	88.32	88.40	-2.03
Middle-heavy	M→I	80.04	79.48	76.89	88.10	87.19	87.26	+0.08
Late-heavy	I→M	80.04	77.63	74.49	88.10	87.49	86.88	-3.14
Late-heavy	M→I	80.04	80.41	78.19	88.10	86.13	86.20	+0.08

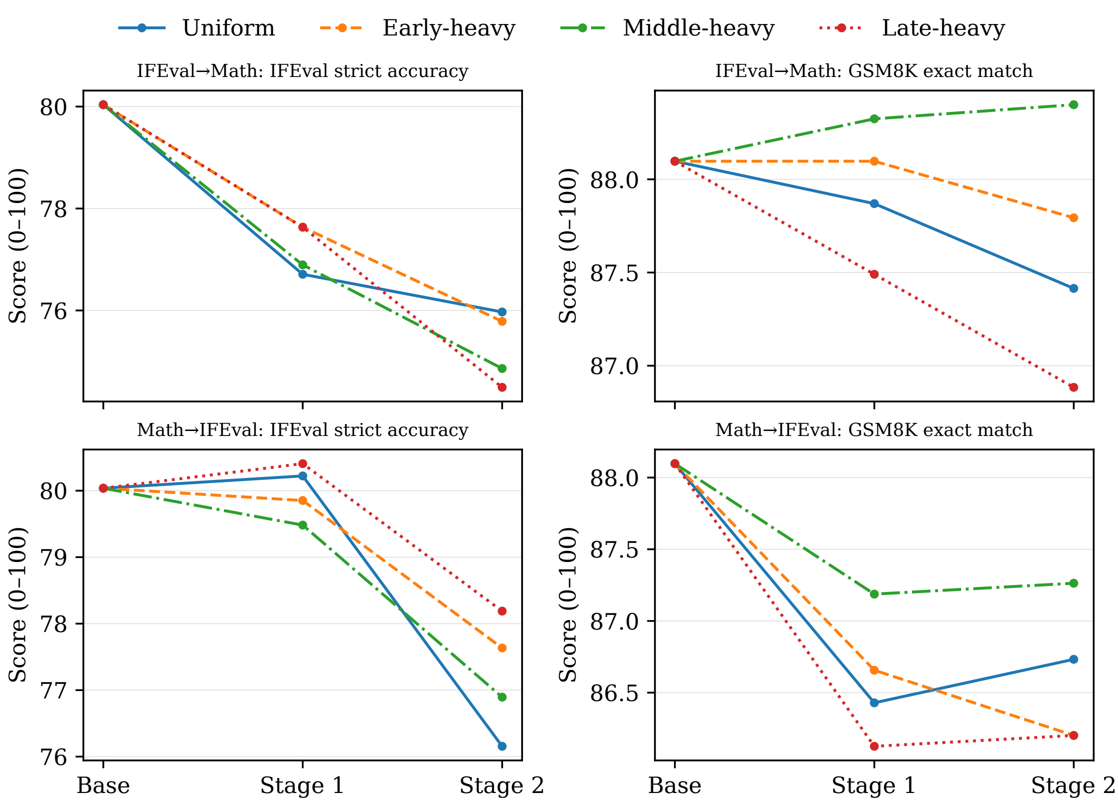

Figure 2. Absolute benchmark trajectories. Stage 1 follows the first task in the named order; stage 2 follows the second. Axes are independently focused within panels, so numerical comparisons should use Table 2.

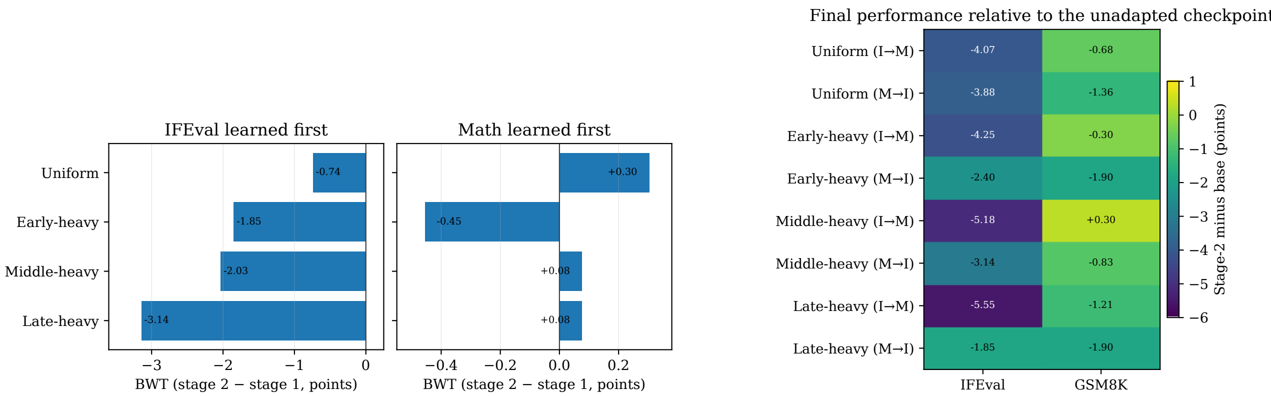

Figure 3. **Left:** first-task backward transfer. Instruction-following retention is more rank-sensitive when mathematics is learned second; math-first interference is small and sometimes positive. **Right:** final score minus the unadapted checkpoint; middle-heavy IFEval→Math GSM8K is the only positive cell.

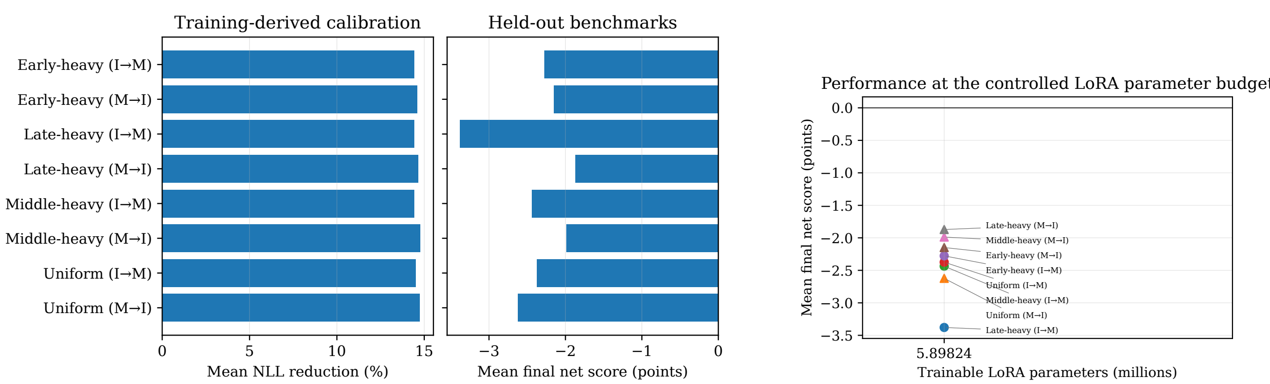

Figure 4. **Left:** every adapter improves assistant-token NLL on the decontaminated, training-derived calibration splits, while mean external benchmark net performance is negative; the panels use distinct units. **Right:** mean final net score at the common 5.898M-parameter budget; every point has the same true horizontal coordinate.

C. Spectral Pilot and Automatic Allocation

The separate pilot trains uniform rank 32 on an alternating, equal-token mix of the two decontaminated training sets. For $B = Q_B R_B$ and $A^\top = Q_A R_A$, the nonzero singular values of BA equal those of the small matrix $R_B R_A^\top$. Tests against an explicit SVD verify this identity. We average normalized cumulative energy from `q_proj` and `v_proj` within each block, then allocate increments from $\{4, 8, \ldots, 32\}$ by marginal energy under Equation 1.

The pilot produces the exact-budget vector

$$
\begin{aligned}
&(24, 20, 20, 20, 24, 24, 20, 16, 16, 16, 12, 12,\\
&\quad 16, 16, 12, 12, 12, 12, 12, 12, 12, 12, 12, 16,\\
&\qquad 16, 16, 16, 20, 16, 16, 16, 20, 16, 16, 16, 12).
\end{aligned}
$$

It is non-uniform without a forced tie-break and contains exactly 5,898,240 parameters. Its final two-order sequential evaluation was not completed, so it is excluded from every comparative claim in the main paper. The pilot adapter is never reused for final training.

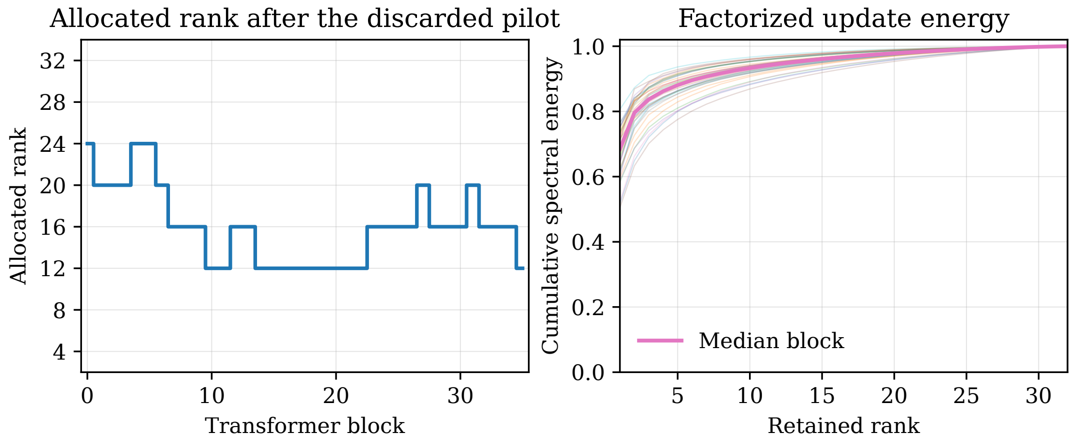

Figure 5. Pilot-only spectral allocation (left) and cumulative energy of the factorized updates (right). Thin curves are block-wise averages over the two target modules; the thick curve is their median.

D. Limitations and Next Experiments

The sweep has one training seed. Full-benchmark evaluation removes sampling noise from prompt subsampling, but not optimization variance; standard deviations and confidence intervals are therefore unavailable. One correct item changes IFEval by 0.1848 point and GSM8K by 0.0758 point, so the smallest gaps should not be overinterpreted. The study covers one instruction checkpoint, two task families, two target projections, and a 4-bit NF4 backbone without a BF16 or full-finetuning control. The high initial benchmark scores also leave limited headroom and may make the post-trained checkpoint sensitive to supervised adaptation. Because the packed training-set sizes are not divisible by four, the standard DDP sampler pads each epoch by one sequence; this exposure is identical across allocations but means physical token presentations slightly exceed the unique prepared-token budget.

The next decisive experiment is three seeds for uniform, middle-heavy (the benchmark-free selected manual candidate), and the spectral allocation in both orders. A higher-precision control would test whether the observed external-benchmark degradation is specific to quantized adaptation. Longer task sequences and the `q/k/v/o` ablation would then test whether the order-dependent pattern generalizes beyond the present setting.

E. Artifact Generation

From the repository root, the audited paper snapshot and all vector figures are regenerated with:

```
uv run python scripts/analyze_results.py \
  --results_dir results \
  --config_dir configs
uv run python \
  scripts/build_first_paper.py
```

The builder rejects any missing protocol, non-scientific manifest, stale configuration fingerprint, incomplete benchmark, unequal parameter budget, or unexpected token budget. The generated CSV and JSON snapshot under `paper/data/` provide the exact inputs used by this manuscript.